WALKWISE

A Data-Driven Urban Design Evaluation Tool for Assessing Walkability Using 5D Criteria

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ABSTRACT
Walkability enhances quality of life, sustainability, and social interaction by reducing car reliance and promoting pedestrian activity. Traditional assessments often use fragmented data, limiting their effectiveness. This study integrates urban morphology and walkability metrics using the 5D criteria—Density, Diversity, Design, Destination Accessibility, and Distance to Transit—to establish a transparent, data-driven framework. Evaluated on a case-study in Copenhagen, the tool evaluates urban designs with a replicable, systematic approach. Applied to Jernbanebyen, a redeveloping project in the western district of Copenhagen, it benchmarks shortlisted proposals from COBE, BIG, and Snohetta, enabling iterative improvements through data-driven insights. This approach equips planners with a systematic method to enhance walkability and urban sustainability. Walkwise can be operated at https://www.walkwisecopenhagen.com/.

1. INTRODUCTION

Urban design faces challenges in developing systematic and transparent approaches to address the complexities of modern cities. This thesis contributes to this challenge by proposing a novel walkability index tool that integrates urban morphology with amenities accessibility, conceptualized as Human Activities Measurements. The tool aims to assess walkability though 5 comprehensive parameters, bridging data collection and analysis, and subsequent design processes, to better align urban form, and development decision making with human behaviours and preferences.

Applied to Copenhagen's Jernbanebyen neighbourhood, a former railway district envisioned as a car-free residential development, Walkwise benchmarks various districts in the city fabric and evaluates the shortlisted urban redevelopment competition proposals from architecture practices COBE, BIG, and Snohetta. Deployed as a friendly User-Interface via Mapbox API, Walkwise provides feedback on design proposal based on quantifiable walkability metrics, enabling data-driven design refinement. The tool also enhances spatial visualization and promotes public participation, allowing non-specialist residents to engage directly in the urban planning process and collaborate with designers to shape walkable neighbourhoods.

By integrating quantitative analysis with participatory spatial visualization, Walkwise offers a practical and transparent approach to evaluating urban walkability, contributing to more inclusive, data-informed, and walkable urban environments.

2. RESEARCH BACKGROUND

The concept of walkability has its roots in Kevin Lynch's work The Image of the City (1960), which highlighted the importance of human-scale urban design. Jane Jacobs (1961) later emphasized the value of "walkable neighbourhoods" in her influential book The Death and Life of Great American Cities. More recently, Jeff Speck (2012) helped popularize walkability as a key metric for evaluating urban quality in his book Walkable City. In recent years, tools like Walk Score (Walk Score, n.d.) and TU Delft's Mapping Walkability: Amsterdam Edition (Kandt & Bertolini, 2021) have introduced data-driven approaches to evaluating walkability, focusing on accessibility and transit isochrones, and incorporating participatory methods to create city-specific indices.

Building on these advancements, this thesis proposes a novel approach that integrates machine learning and interactive spatial visualization. A comprehensive tool has been developed to assess walkability through the lens of urban morphology, applying the 5D criteria—Density, Diversity, Design, Destination Accessibility, and Distance to Transit—to offer a more nuanced, dynamic, and scalable framework for analysing walkable environments.

3. AIMS AND FRAMEWORK OF THE STUDY

The workflow begins with a Literature Review, followed by site analysis to gather relevant data. This involves open datasets, and municipal records to capture relevant features such as street networks, land use patterns, and transit infrastructure. The gathered data is then standardized and processed to construct the key walkability Indicators, where each of the 5D variables is quantified.

For Urban Proposal Evaluation, design proposal from Cobe, BIG, and Snøhetta are analysed by converting Rhino DXF files into GIS-compatible formats and reprojecting them into a consistent Coordinate Reference System (CRS). This allows for direct comparison between existing urban conditions and proposed interventions. Walkability is assessed at both the city scale (macro-level trends) and proposal scale (site-specific impact), applying the 5D Framework to evaluate how each design performs against baseline conditions.

The proposed web-based platform operationalizes data-driven urban design by implementing the 5D walkability framework (Density, Diversity, Design, Destination Accessibility, Distance to Transit) through advanced computational methods. By processing urban proposals via Graph Theory for network analysis, Machine Learning for dealing with missing data, and Mapbox API for geospatial visualization, the system generates quantifiable metrics including block-wise amenity distances, street edge classifications, and composite walkability indices. These analyses are synthesized into an interactive web interface that enables comparative evaluation of design scenarios against baseline conditions, featuring dynamic visualizations such as isochrones and network graphs.

The platform facilitates iterative design refinement through real-time metric updates, allowing both planners and citizens to assess trade-offs and optimize proposals. As illustrated in Figure 01, this end-to-end workflow bridges technical analysis with practical decision-making, transforming raw spatial data into actionable insights for sustainable urban development.

4. METHODOLOGIES

4.1. DATA COLLECTION, TOOL DEVELOPMENT AND DEPLOYMENT

We employ graph-based spatial analysis techniques in Python to investigate urban networks at both district and neighbourhoods' scales. Using various spatial methods—such as bounding boxes, pixel-based grids, buffer zones, and shortest-path algorithms—we assign and transform values across spatial elements including nodes, edges, and grids. Each method serves a distinct purpose: bounding boxes are used to calculate tree coverage and local travel times; pixel/grid-based analysis supports ratio calculations and final benchmarking; buffer zones enable localized computation of floor area ratio (FAR); and the Haversine distance algorithm is applied to estimate travel times between key urban destinations /features. (Figure 02)

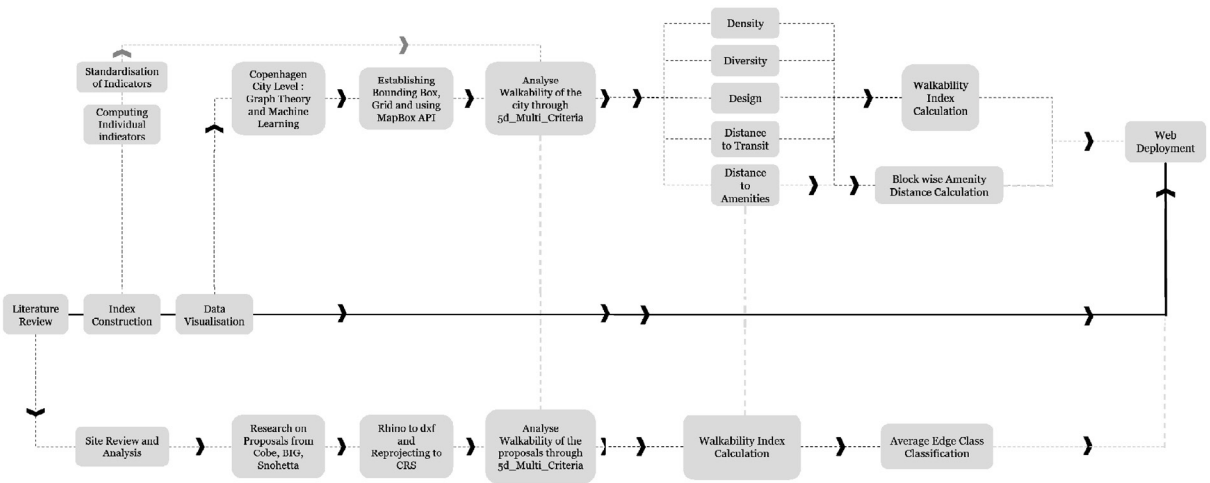


Figure 1:
Workflow for walkability analysis and evaluation

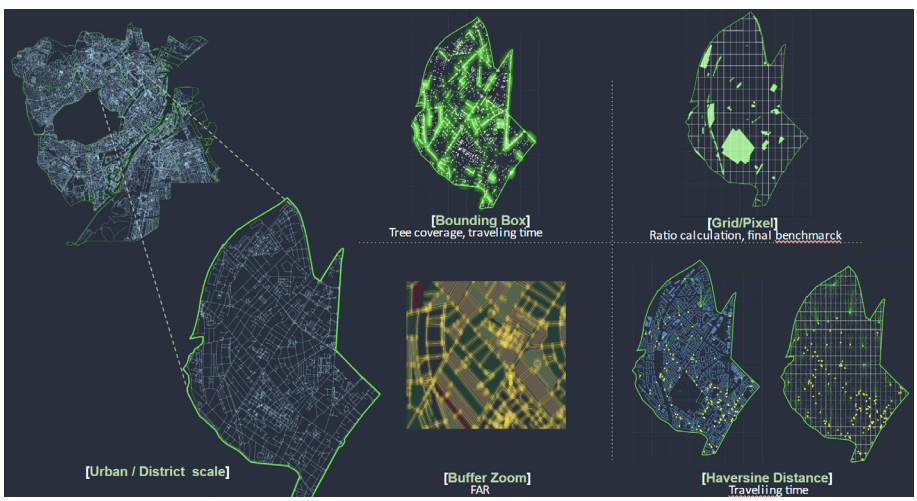


Figure 2:
Graph-Based analysis of urban networks

4.2. ANALYSIS WORKFLOW

4.2.1. URBAN SCALE ASSESSMENT

To assess walkability at the urban scale, we apply the 5D framework—Density, Diversity, Design, Destination Accessibility, and Distance to Transit—across the city of Copenhagen. Each of these dimensions captures a key aspect of the built environment that influences pedestrian experience. The analysis is conducted using Mitchell's grid tessellation method (Figure 03), generating uniform spatial units at varying resolutions (100 × 100 m to 500 × 500 m). Within each grid cell, spatial indicators are calculated:

a. Distance to Transit (figure 04):

- Transit proximity (e.g., distance to nearest stop and metro station)

b. Destination Accessibility (figure 05):

- Proximity to essential services
- Amenity density based on OpenStreetMap (OSM) data

c. Density (figure 06):

- Residential Population density
- Street intersection density
- Floor Area Ratio (FAR)

d. Diversity (figure 07):

- Mix of landuse

e. Design (figure 08):

- Green space coverage
- Tree density
- Green ratio
- Block Compacity

Figure 3:
Mitchell's grid

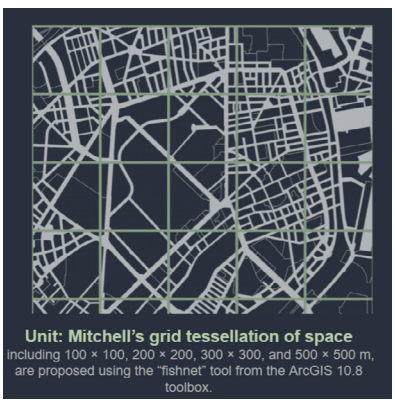
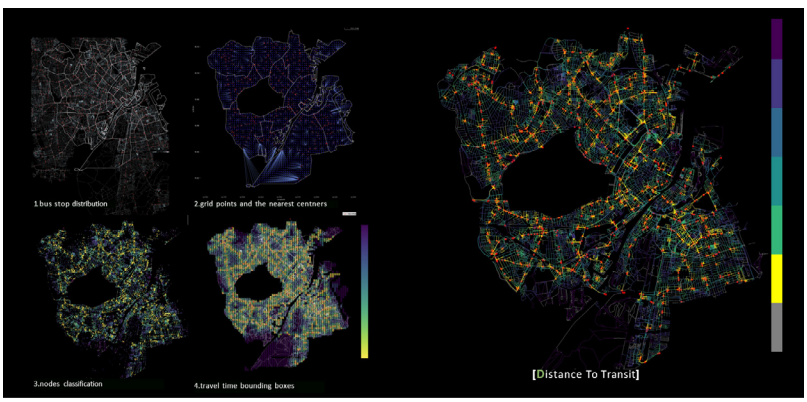


Figure 4:
Distant to Transit . We use all bus stop locations as destination points and apply the same methodology, resulting in a well-structured network edge graph.



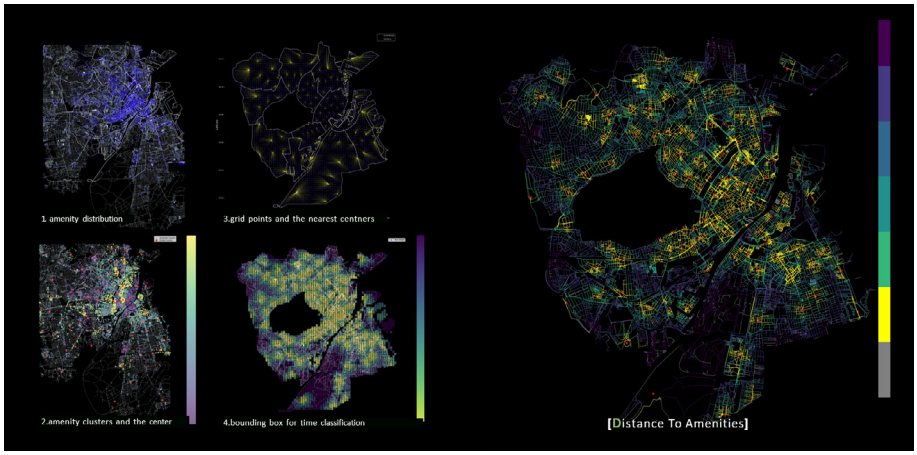


Figure 5:
Destination Accesibility. Identify all amenity locations, determine the cluster centers, and create a matrix to categorize travel distances for each location.

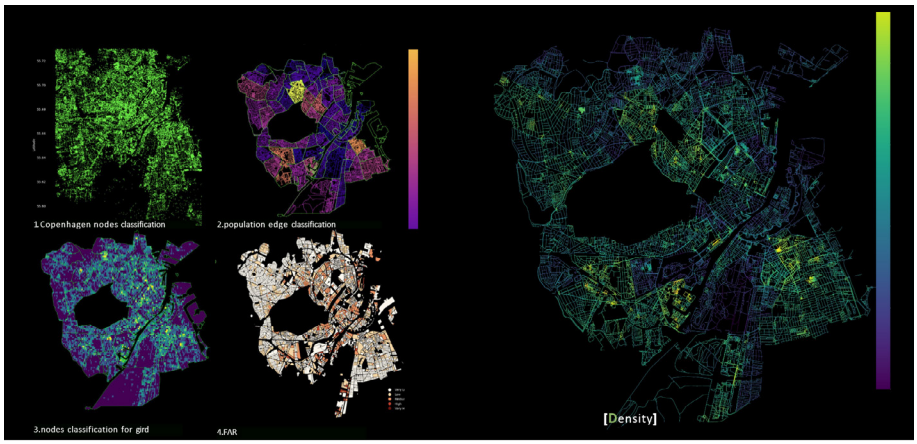


Figure 6:
Density. The density calculation incorporates multiple factors, including node density, population data, and Floor Area Ratio (FAR), to provide a comprehensive measure of urban density.

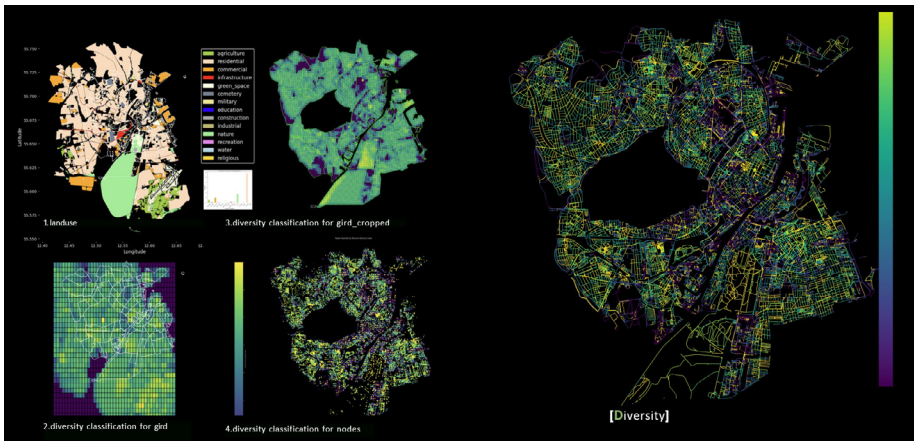


Figure 7:
Diversity. The diversity component in walkability calculations refers to the variety of land uses within an area.

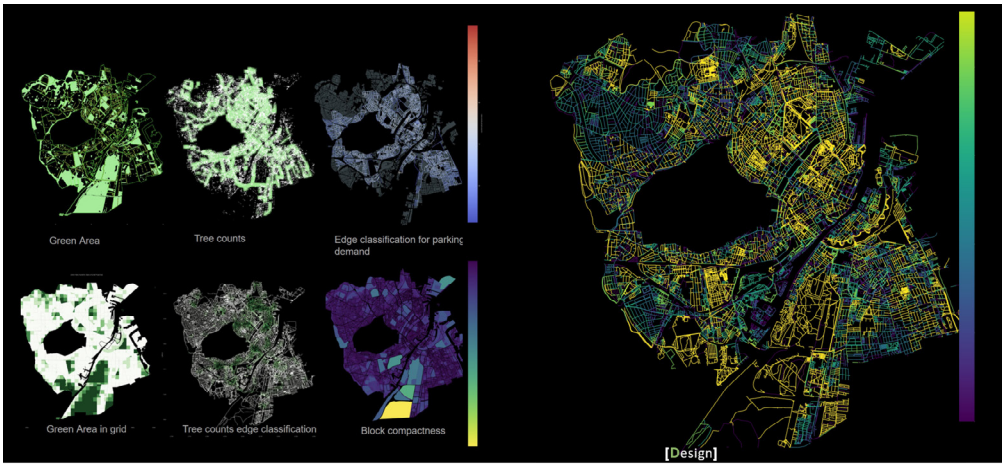


Figure 8:
Visualizing the Design component of the Urban Scale Assessment.This includes street tree count, green space ratio,parking demand, and block compactness.

4.2.2. MACHINE LEARNING APPLICATIONS

To assess walkability at the urban scale, we apply the 5D framework—Density, D Graph machine learning is a subfield that uses graph structures (Figure 09) to analyse and predict relationships between interconnected data points, employing techniques like graph neural networks.

During our analysis, we identified missing data (Figure 10) for Distance to Bus, Metro, and Amenities, prompting us to apply graph machine learning's transudative edge classification to fill these gaps.

Accurate graph machine learning relies on high-quality datasets, including the road network (classified into primary, secondary, tertiary, and residential), bus stops, metro stations, amenities, and grid point classes. Using OSMNX, the city graph is built with features assigned to nodes and edges from these datasets. Edge classification predicts missing edge classes by utilizing node features within a defined neighbourhoods radius to inform edge properties, enhancing data accuracy and improving model reliability.

After assigning features to nodes(Figure11.), known classes are applied, and nodes with missing classes are identified. The graph is converted to DGL for graph machine learning, focusing on edge classification. The model achieves 70–80% accuracy, with validation and training curves showing consistent performance without overfitting. The confusion matrix indicates accurate predictions for classes 1–4, though classes 5 and 6 are less reliable.

This accuracy is sufficient for predicting distances to amenities or transit. The final analysis highlights that Copenhagen has strong transit infrastructure, but accessibility to amenities diminishes further from the city centre (Figure12).

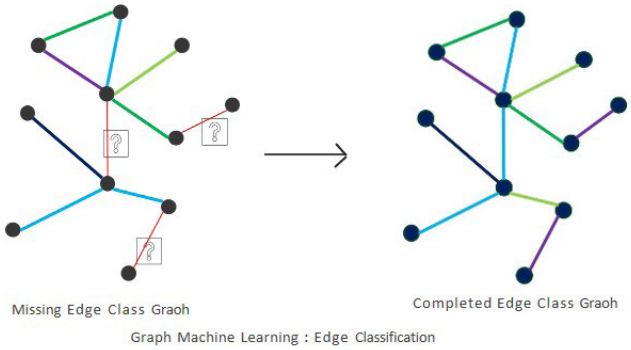


Figure 9:
Graph-based node feature assignment and classification results

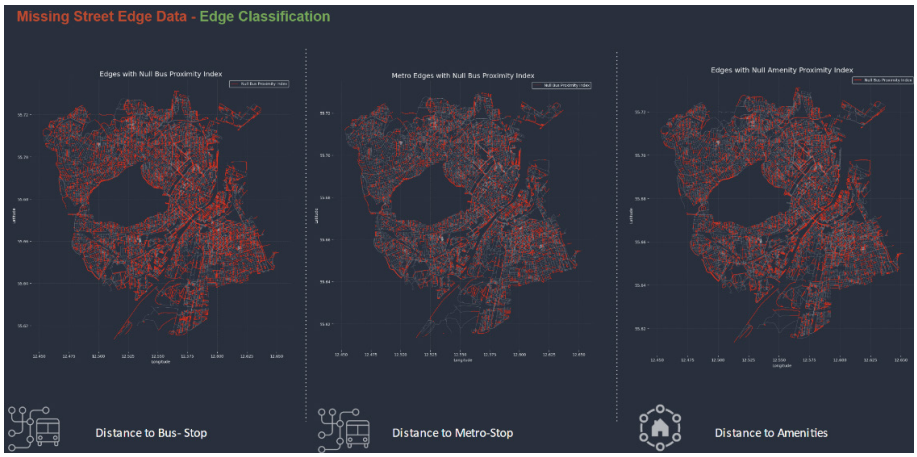


Figure 10:
Edges graph with null amenity proximity

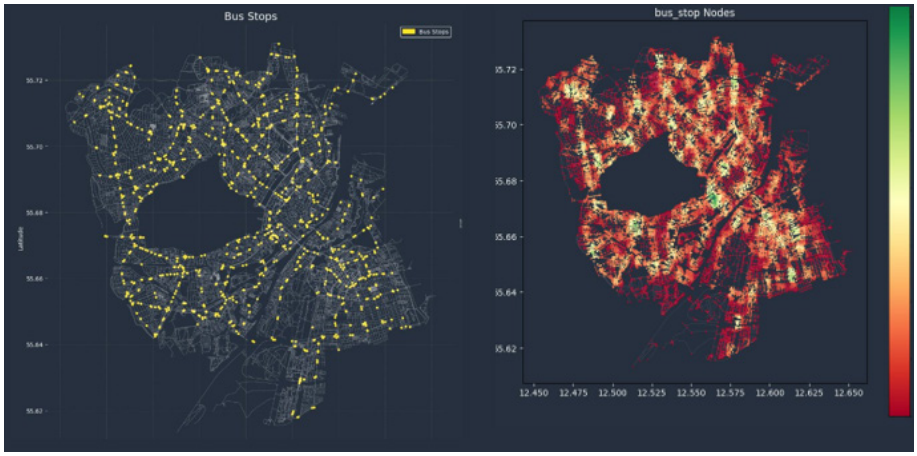


Figure 11:
Location points and assigned node feature graph

4.2.3. BLOCK FORMAT FOR DESTINATION ACCESSIBILITY DISPLAY

The analysis uses Copenhagen block data, population data, and OSM amenities to measure distances from block centroids to services. A dynamic map visualizes walkability, shading blocks by walking time (Figure 13).

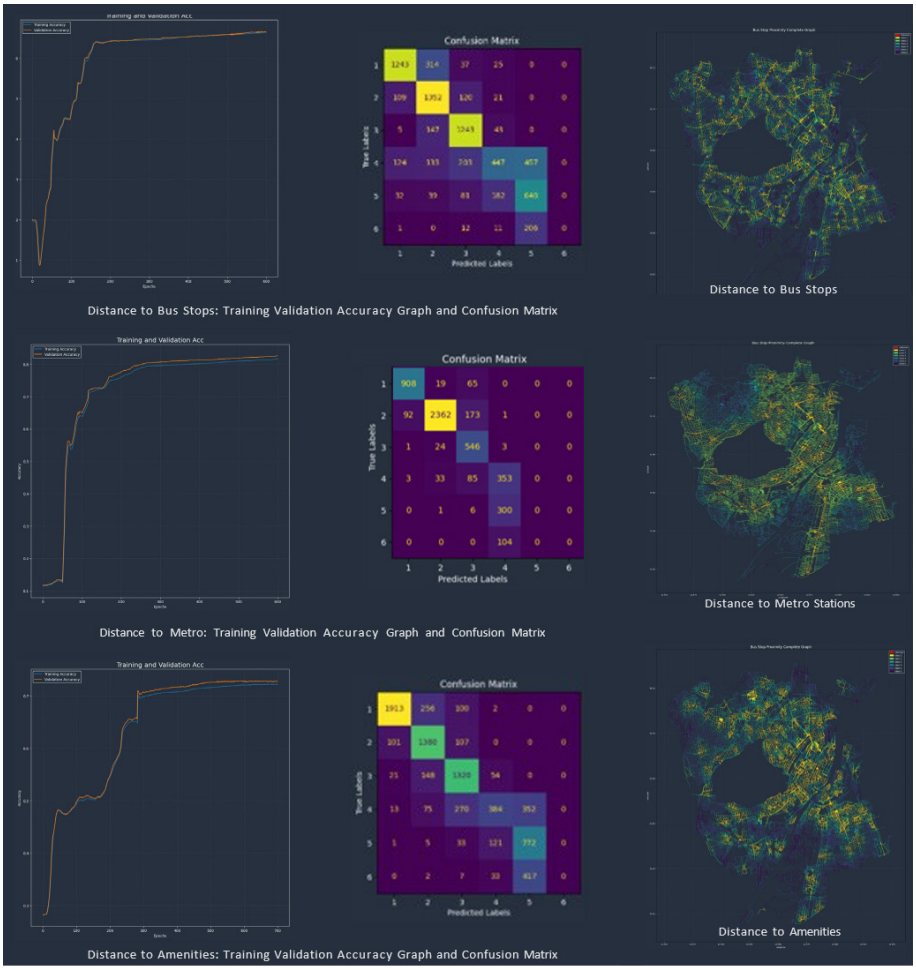


Figure 12:
Graph-based node feature assignment and classification result

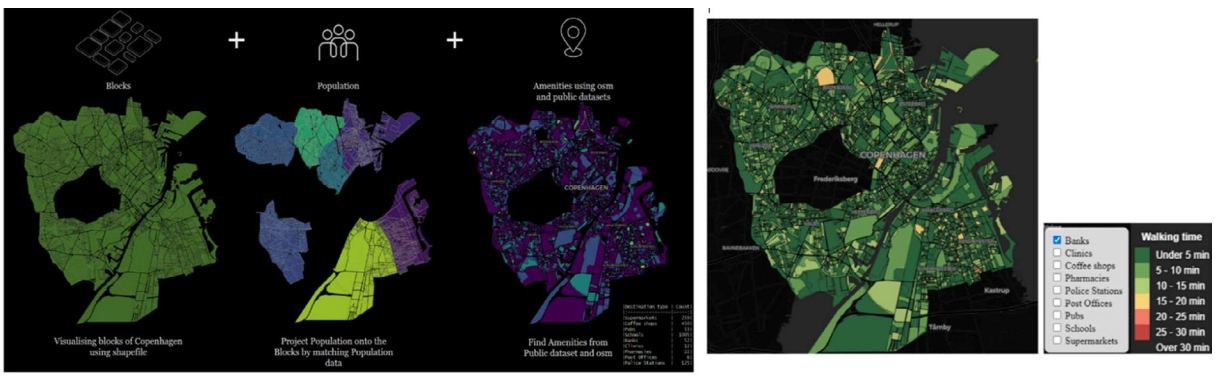


Figure 13:
Distance to amenitis and highlights walking distances to key amenities, visualizing accessibility across the city.

4.2.4. URBAN PROPOSAL EVALUATION/RHINO TO GRAPH WORKFLOW

To enable a comprehensive walkability analysis, urban design proposals created in Computer-Aided Design (CAD) software such as Rhino or AutoCAD must be converted into geospatially analysable formats.

This study follows a structured workflow to transform these CAD-based urban proposals into spatial data compatible with walkability metrics and spatial network analysis (Figure 14).

Step 01: Data Structuring in Rhino and AutoCAD

To ensure consistency and analytical usability, CAD files must adhere to a predefined organizational framework before being exported to DXF format (Table 01). This framework includes:

- Layer Organization: Design elements are categorized into standardized layers—e.g., Buildings, Trees, and Land Use—with sub-layers providing further granularity (e.g., Residential and Commercial under Buildings).
- Naming Conventions: A strict naming system facilitates automated processing and accurate attribute extraction.
- Line Type Encoding: Specific line types encode essential attributes such as building height, street type, or pedestrian path classification.
- Spatial Projection: Files originally in Cartesian coordinates are prepared for conversion to a geographic Coordinate Reference System (CRS), typically WGS 84 (EPSG:4326), for global compatibility.

Step 02: DXF to GeoDataFrame (GDF) Conversion

Using QGIS, DXF files are imported and real-world coordinates are assigned by referencing auxiliary geospatial datasets. This ensures accurate geospatial alignment of buildings, streets, and open spaces within the existing urban fabric.

Step 03: Georeferencing Using QGIS
Georeferencing transforms the CAD drawing's local coordinates into geographic coordinates. Reference layers (such as OSM basemaps or municipal datasets) help align design elements to their correct spatial locations, ensuring interoperability with other datasets.

Step 04: Translation and Scaling
Scaling factors and translation adjustments ensure accurate spatial alignment. These refinements integrate multiple urban proposals within a city-scale context, making comparative analysis possible.

Step 05: Projection to CRS

For full interoperability, spatial data is re-projected from Web Mercator (EPSG:3857) to WGS 84 (EPSG:4326). This allows seamless integration with global datasets such as OpenStreetMap, census data, or environmental models (Figure 15).

Step 06: Updating the GeoDataFrame
The Geodata Frame is enriched with new attributes, including land use classifications, building footprints, and distances to transit stops.

These additions enhance the potential for multidimensional spatial analysis.

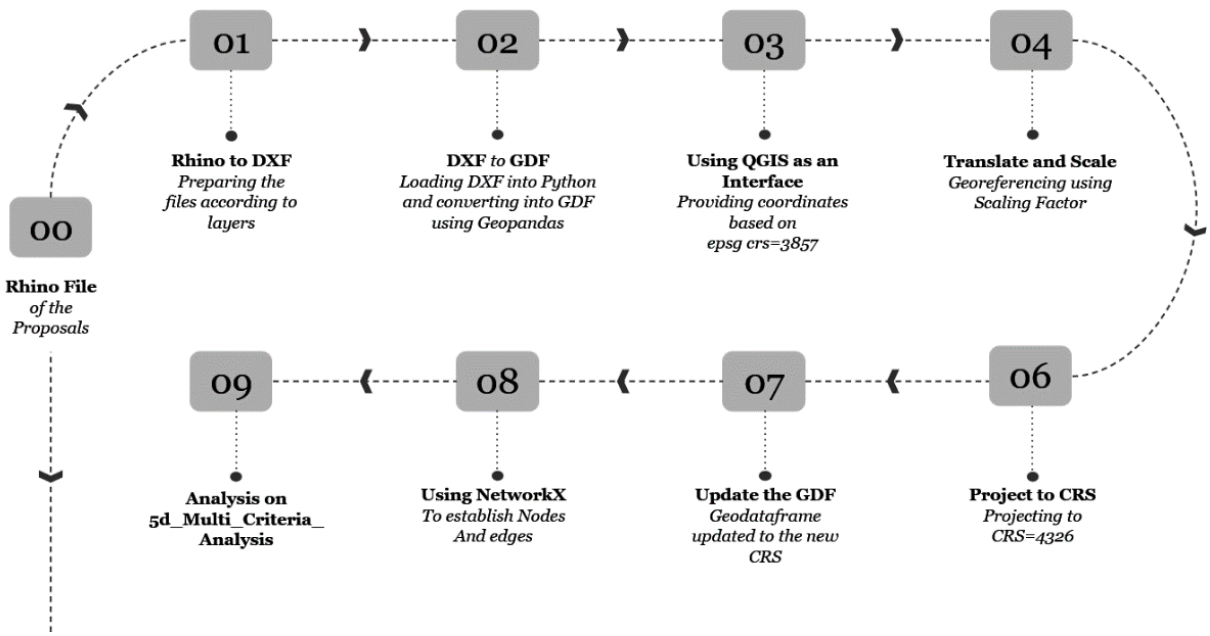
Step 07: Network Graph Construction
Once the spatial data is structured, NetworkX establishes nodes (intersections, amenities, buildings) and edges (streets, pedestrian paths), forming a graph representation of the urban environment (Figure 15). The resulting network graph enables the computation of key metrics, such as betweenness centrality, connectivity indices, and shortest-path distances, providing quantitative insights into pedestrian accessibility.

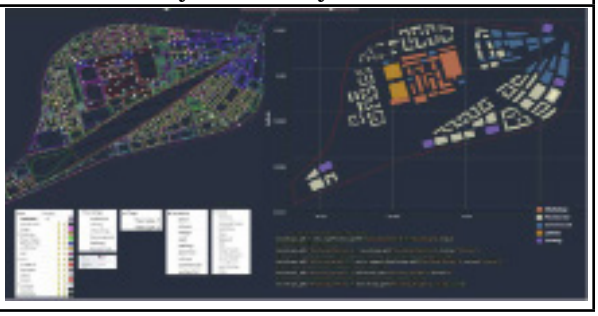
Step 08: 5D Walkability Analysis
The network is analysed using the 5D walkability framework, computing key metrics: Density, Diversity, Design, Distance to Transit, Distance to Amenities. These indicators help assess the pedestrian-friendliness of urban layouts, identifying areas where improvements can enhance walkability. The inclusion of real-world pedestrian movement data, when available, can further validate these results.

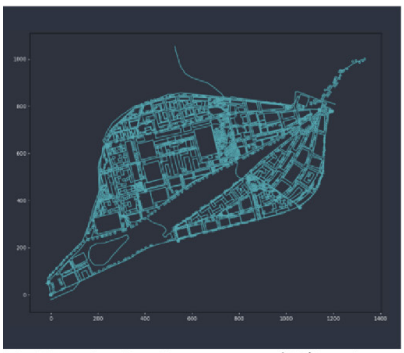
This structured workflow enables the seamless transformation of CAD-based urban proposals into analyzable geospatial datasets. By combining spatial data processing with network graph analysis, the methodology supports a detailed assessment of pedestrian connectivity, accessibility, and urban form.

The use of standardized formats and projections ensures interoperability with a wide range of analytical tools and datasets. Moreover, the framework supports scenario-based simulations—allowing planners and policymakers to evaluate the potential impact of design modifications on pedestrian mobility and urban quality.

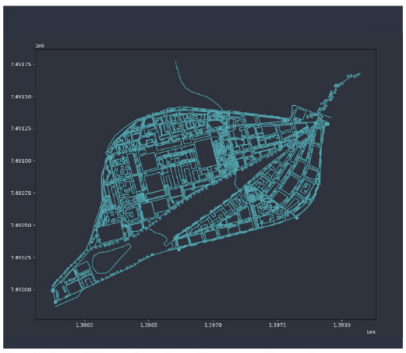
Figure 14:
Rhino to graph workflow



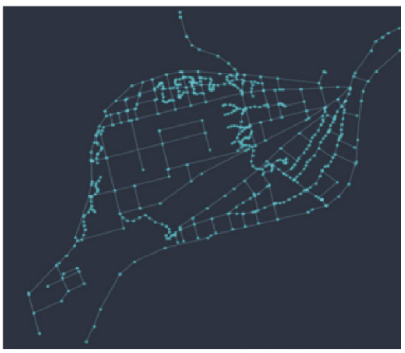
	Masterplan Layout	From Dxf to Python Analysis
Cobe		



Inital Cartesian Coordinate system at(0,0) location



Scaled and Translated to the exact Location on map



Converted to nodes and edges for further analysis

Table 1:
Setting Up the dxf File for Analysis

Figure 15:
Converting Cartesian Coordinates to WGS 84 and Creating Nodes and Edges

4.2.5. ASSIGN AMENITIES

Due to missing amenities data in masterplans, this method assigns amenities based on the 15-minute principle, ensuring walkable access to essential services and promoting sustainability.

4.2.5.1. AMENITY DEMAND PROFILES

Amenity factors, analysed with Python (Figures 16 & 17), allocate amenities based on population needs, aligning with 15-minute city principles to optimize urban planning in Jernbanebyen.

The data supports sustainable living by integrating amenities based on population needs. Visualized via Grasshopper and Rhino (Figures 18), it categorizes amenities for efficient urban planning.

```
# Define activity demand profiles
demand_profiles = {
    'Residential': {'Local Shops': 0.4, 'Health Clinics': 0.3, 'Community Centers': 0.2, 'Pharmacies': 0.1},
    'Commercial': {'Shops': 0.4, 'Restaurants': 0.3, 'Cafes': 0.2, 'Medical Facilities': 0.1},
    'Workshop': {'Co-working Spaces': 0.5, 'Training Centers': 0.3, 'Local Workshops': 0.2},
    'Library': {'Reading Rooms': 0.5, 'Study Areas': 0.3, 'Digital Resource Centers': 0.2}
}
```

Figure 16:
Defining amenity factors through Python by allocating necessary percentage based on 15 minute city principle

```
# Define different amenity factors for each area based on 15-minute city principles
amenity_factors = {
    'Residential': 0.07,  # 7% of the population will be amenities
    'Commercial': 0.12,   # 12% of the population for commercial areas
    'Workshop': 0.05,     # 5% of the population for workshops
    'Library': 0.03       # 3% of the population for libraries
}
```

Figure 17:
Defining Activity demand profiles within the amenity factors allocated based on 15 minute city principle



Figure 18:
Amenities allocated using Grasshopper workflow for Proposal by Cobe Architects.

5. RESULTS

5.1. WALKABILITY INDEX CALCULATION

The multi-criteria framework evaluates Density, Diversity, Design, Destination Accessibility, and Distance to Transit, tailored to Copenhagen based on specifically adapted to the Copenhagen context through a custom-built research database (Figure19):

- Density: Population, node intersections and FAR.
- Diversity: Mixed land use and service distribution.
- Design: Street connectivity, public spaces, and pedestrian infrastructure.
- Destination Accessibility: Proximity to key locations via shortest-path analysis.
- Distance to Transit: Pedestrian access to public transportation

The result is merged layout from the 5D walkability component for whole Copenhagen (Figure 20), which are visualized on an interactive web interface built with Mapbox API.

5.2 BENCHMARKING URBAN PROPOSALS

The study evaluates three urban proposals (BIG, Cobe, Snøhetta) for Copenhagen's Jernbanebyen district using a data-driven 5D. Key findings:

a. **Benchmark Design:** Cobe's proposal excels with centralized public spaces and balanced tree density , while Snøhetta shows higher block compactness (Figure 21).

b. **Benchmark Density:** Snøhetta's plan clusters high-density nodes (avg. 3.88) near pathways, enhancing connectivity (Figure 22).

c. **Benchmark Accessibility:** Cobe ranks highest in proximity to amenities and transit (Figure 24), reducing average travel distances (Figure 23).

d. **Benchmark Distant to Transit:** Since only COBE's proposal included detailed information on bus and metro station locations, this dataset was used as a common reference across all three proposals (Figure 24).

e. **Benchmark Diversity:** Mixed land-use intersections boost walkability, with Cobe achieving the most integrated grid patterns (Figure 25).

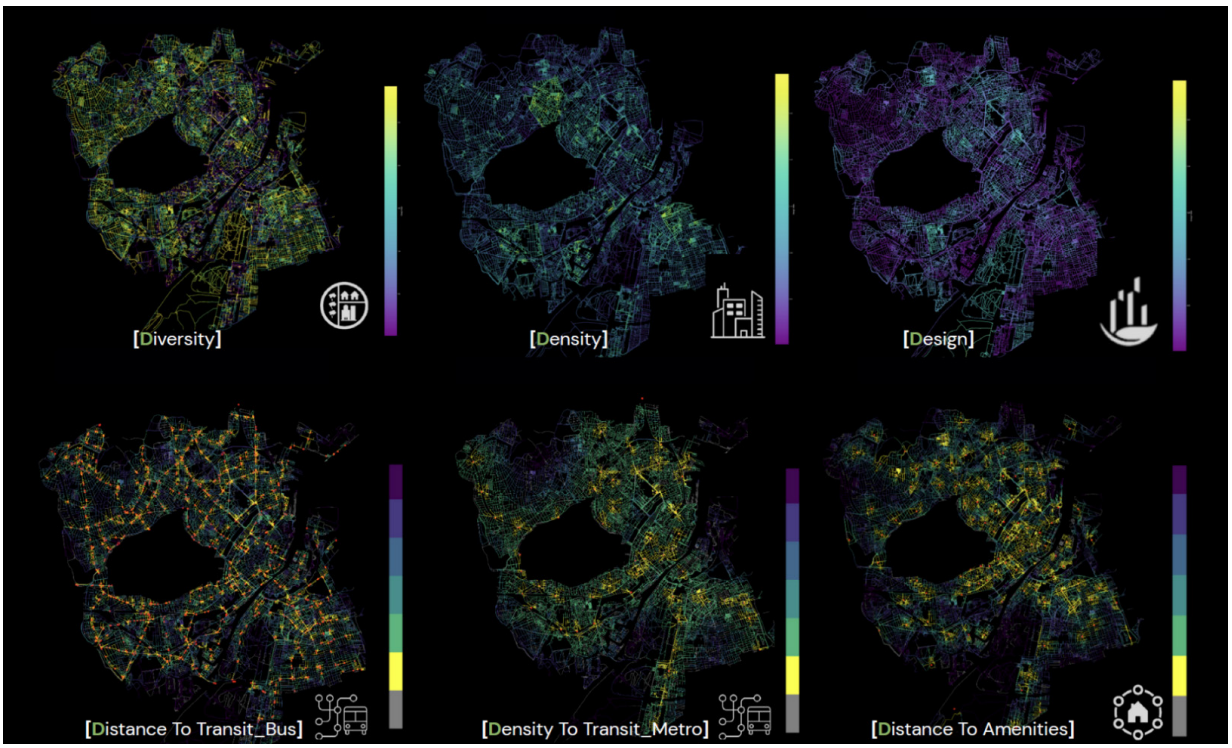


Figure 19:
5 D Walkbility Index calculation and graph analysis

The results indicate that COBE has the highest walkability score among the three proposals, as shown by the grid cell classifications and the histogram of edge classes.

To synthesize the benchmark findings, the table (Table 02) below compares the three urban proposals across the five walkability dimensions, identifying the leading approach in each category.

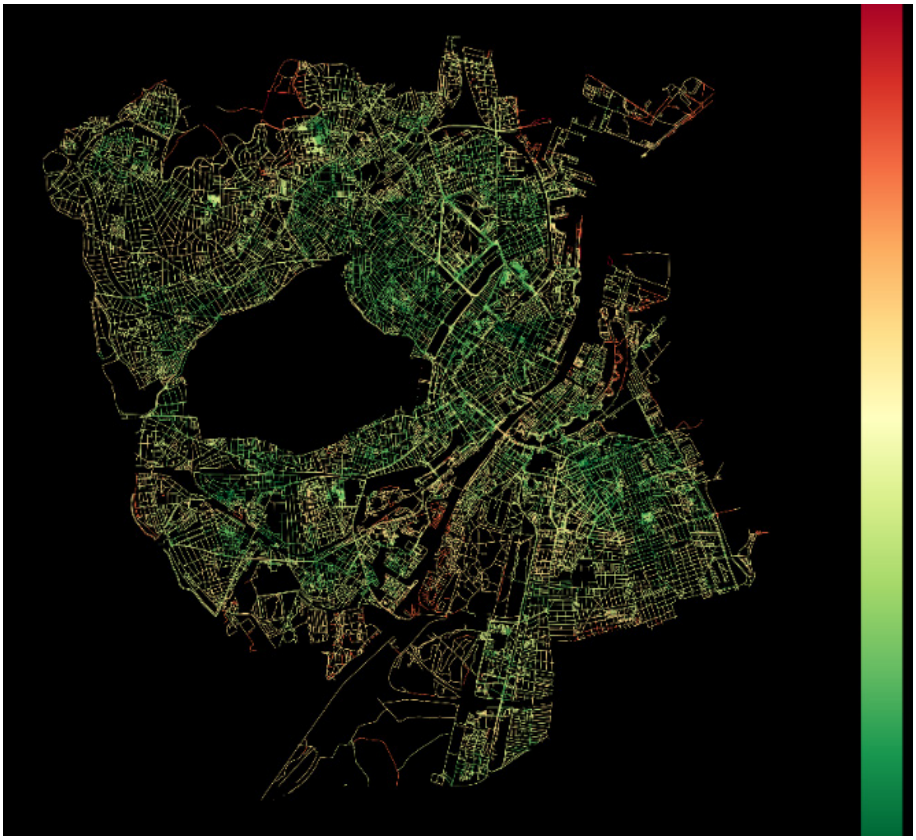


Figure 20:
Final Walkability Index from 1 to 5 for Copenhagen

	BIG	Cobe	Snøhetta	Best Performer	Key Insight
Design	Moderate public space integration	Centralized, vibrant public hub (Fig. 21)	High block compactness	Cobe	Cobe's design fosters community interaction.
Density	Lower density (avg. 3.30)	Balanced density (avg. 3.51)	High-density clusters (avg. 3.88) (Fig. 22)	Snøhetta	Snøhetta's density may reduce walkability if overly clustered.
Diversity	Standard land-use mix	Optimal land-use intersections (Fig. 23)	Moderate mixing	Cobe	More intersections = better walkability.
Distance to Amenities	Longer average distances	Shortest distances (Fig. 24)	Moderate proximity	Cobe	Proximity reduces pedestrian travel time.
Distance to Transit	Less optimized	Best metro/ bus access (Fig. 25)	Partial coverage	Cobe	Transit access critical for urban mobility.
Final Walkability Index	Lower score	Highest score	Intermediate score	Cobe	Cobe's holistic approach wins overall.

Table 2:
Summary of 5D Benchmark Results Across the Three Urban Proposals

The final walkability index (Figure 26) crowns Cobe's proposal as optimal, outperforming BIG and Snøhetta. When benchmarked against Copenhagen's Nørrebro district (Figures 26),,Jernbanebyen's design shows superior walkability (avg. score 2.73 vs. 1.34), aligning with decarbonization goals through pedestrian-centric planning.

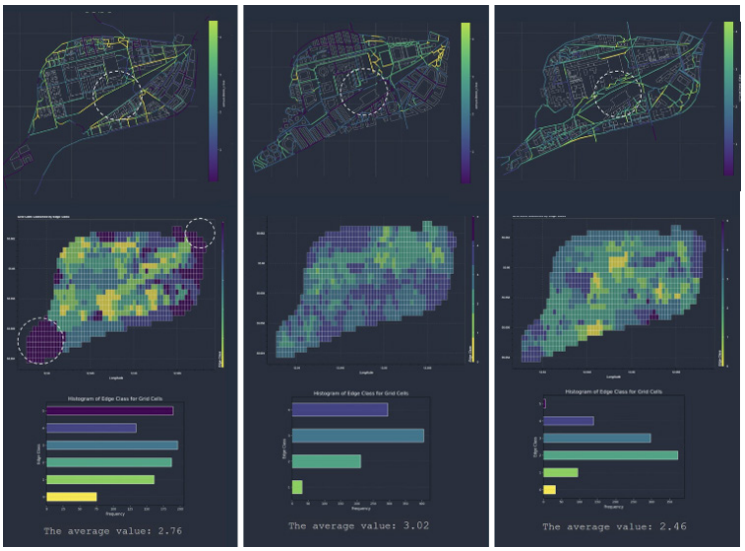


Figure 21:
Benchmark design component. All components are arranged in a pixelated form for easier classification and comparison.

From left to right: Cobe, Big, Snohetta.

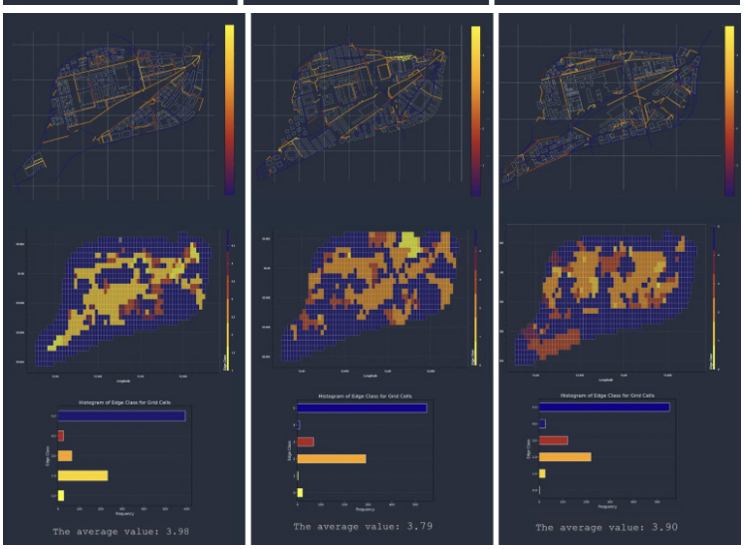


Figure 22:
Benchmark Density component. Grid cells are classified by edge class, with COBE showing balanced density (avg: 3.51), BIG lower density (avg: 3.30), and Snohetta higher density clusters (avg: 3.88).

From left to right: Cobe, Big, Snohetta.

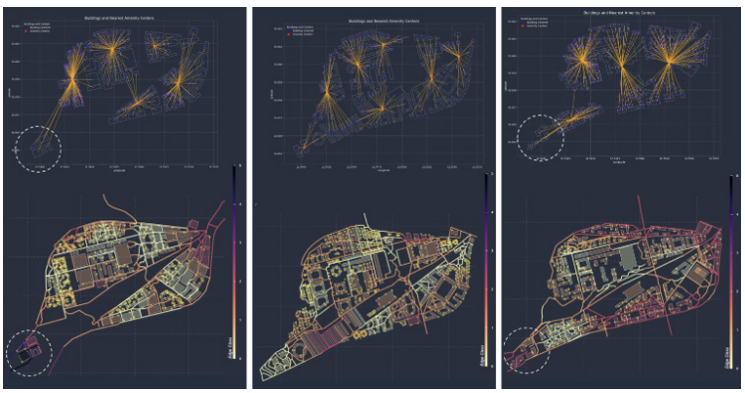


Figure 23:
Benchmark distant to amenities between the 3 proposals.

From left to right: Cobe, Big, Snohetta.

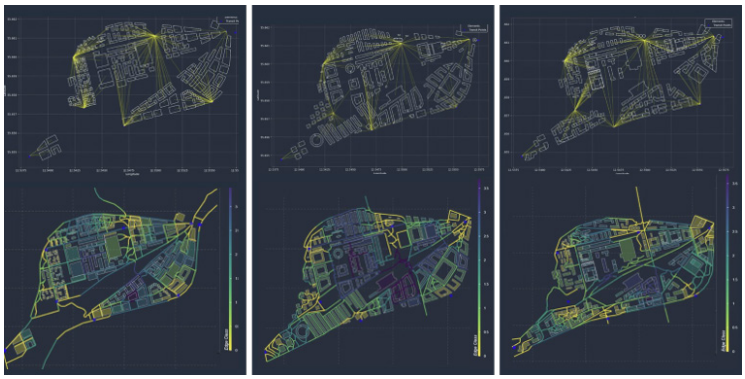


Figure 24:
Distance to Transit. Using bus and metro locations provided by COBE, transit times are calculated for each building to the nearest station.

From left to right: Cobe, Big, Snohetta.

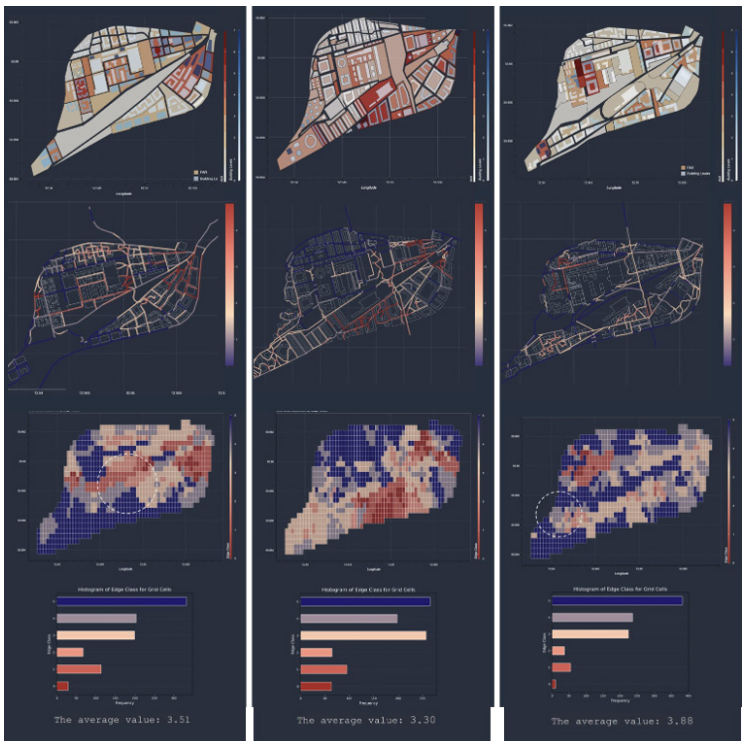


Figure 25:
Benchmark Diversity.
A higher number of intersections between various land-use categories within a single grid significantly boosts its value.

From left to right: Cobe, Big, Snohetta

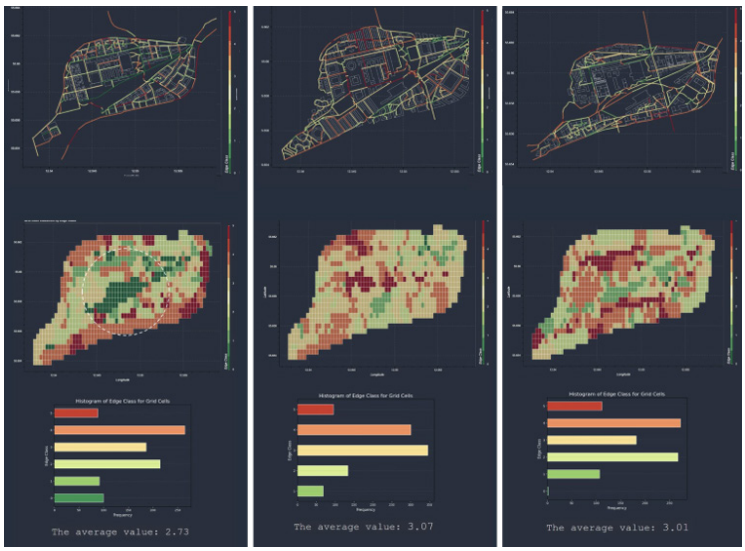


Figure 26:
inal walkability index comparison From left to right: Cobe, Big, Snohetta.



Figure 27:
Overall Walkability in Jernbanebyen and Nørrebro

6. WEB DEPLOYMENT

WalkWise is a web-based tool (Figure 28, Table 03) for walkability analysis, starting with Copenhagen. It evaluates the Walkability Index using the 5D criteria, benchmarking urban proposals against existing conditions pre-construction. The web interface visualizes the index for each block and street. Distance to Transit/Amenities scores range from 0 (good) to 5 (poor), while Diversity, Density, and Design range from 0 (poor) to 5 (excellent). The final index is a standardized average of all five criteria.

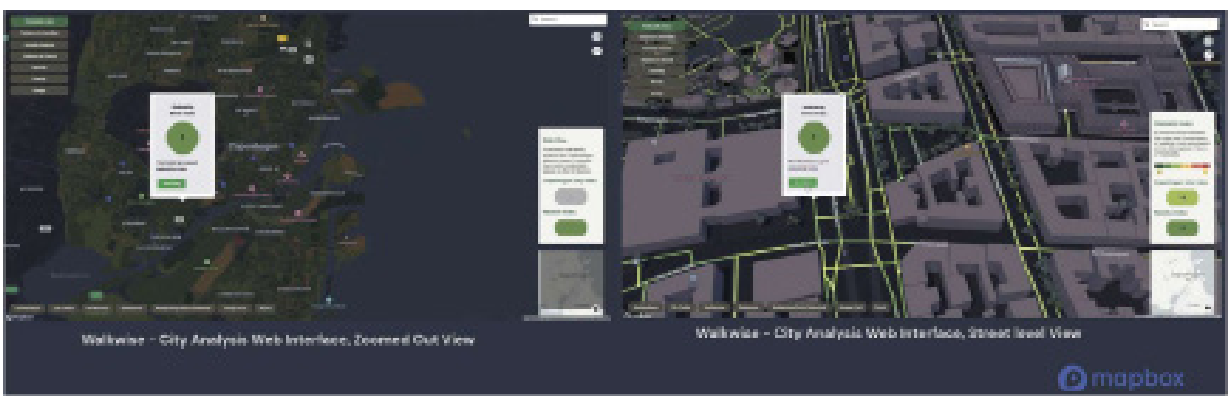


Figure 28:
Urban scale web deployment: Walkwise

The web interface enables uploading and visualizing urban design proposals for Jernbanebyen by integrating Mapbox, Flask, and DXF processing.

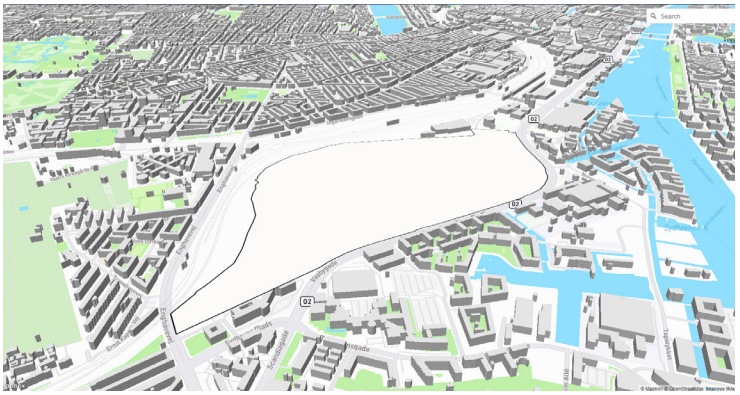
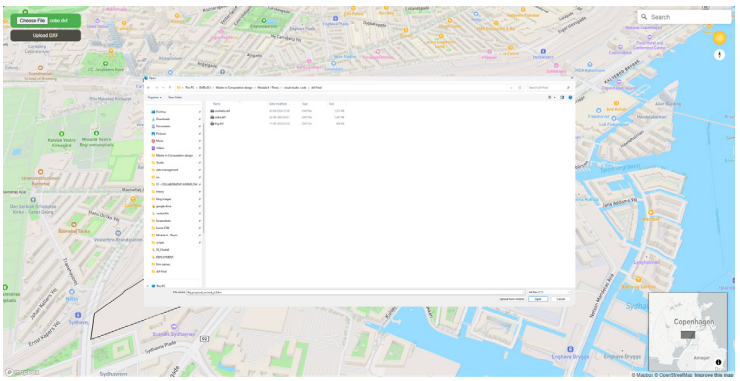
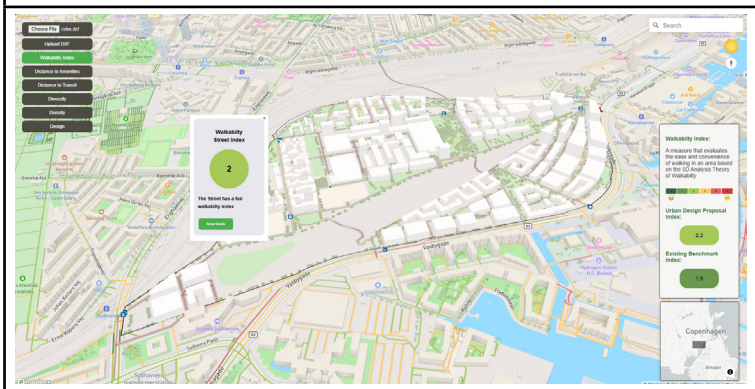
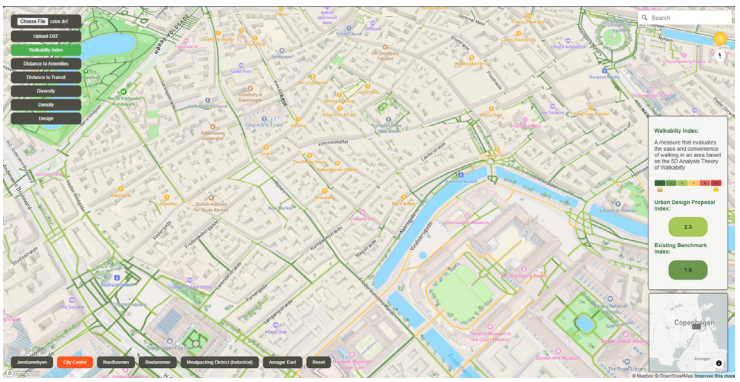
Web Interface	Masterplan Proposal Uploading
	**Cleaning Existing Map:** The Jernbanebyen region have its existing streets and buildings. To upload urban design proposals first we would need to clean the map. The streets and 2d buildings from are made using the geojson layer over it hidding the existing roads since mapbox doesnt support the function to remove existing street segments. The 3D Buildings from the site can be removed with the help of clip layer api which helps erase the 3D buildings in the given area.
	**Choose the DXF:** Next, we start with linking python backend code to HTML using Flask. Once linked we start creating function to load the dxf files. **Upload DXF:** Once the file is selected a upload dxf button is created which when clicked activates the python code and converts the dxf to a model which can be displayed on the map at exact ocation and scale.
	**Anaylsis:** The Python code is similarly linked to the html using flask and similar to city analysis we have info box providing ths analysis index value, popups showing the street index and its details.
	**City Areas:** The toggles at the bottom allow users to quickly navigate to key areas of Copenhagen, helping user easily benchmark the proposals with the existing areas.

Table 2:
Walkwise Web Deployment

7. DISCUSSION OF RESULTS / FINDINGS

The 5D framework provided a data-driven methodology for evaluating walkability in urban design proposals. By employing quantifiable metrics and visual analytics, the tool facilitated objective comparisons and actionable insights for designers and planners. Key findings from this study include:

- Systematic Evaluation: The framework enables a balanced assessment across the five dimensions of walkability, ensuring that urban designs align with human-centric and sustainable urban development principles.
- Interactive Feedback: Visual representations of walkability scores allow stakeholders—including designers, policymakers, and the public—to engage with the analysis and make evidence-based decisions.
- Sustainability and Connectivity: The tool emphasizes pedestrian accessibility, public space quality, and urban connectivity, reinforcing the creation of walkable and environmentally sustainable cities.
- Scalability and Transferability: The methodological approach can be adapted to different urban contexts, offering potential for broader application in global cities striving for improved walkability.

Walkwise was successfully applied to assess urban design proposals for Jernbanebyen, demonstrating its capability to integrate Computer-Aided Design (CAD) models with geospatial data for quantitative walkability analysis. However, scaling the tool to a broader urban or cross-city context presents several challenges:

- Data Accessibility: Urban datasets vary widely in format, quality, and availability across different regions, with some cities lacking fully digitized infrastructure data.
- Data Integration: Harmonizing disparate data sources into a unified analytical framework remains a key technical hurdle.
- Dynamic Data Updates: Urban environments are constantly evolving, necessitating mechanisms for real-time or periodic data updates to maintain accuracy.

8. CONCLUSIONS

This paper has presented a digital evaluation tool Walkwise in order to support the decision making in large-scale urban development. Walkwise defines 5 core parameters which quantify the quality of space within the urban fabric. The tool has been tested against a case study in Copenhagen, where it has evaluated the short-listed proposals.

The deployment of Walkwise through a web-based user-interface argues for the need to democratize decision making by removing entry-barrier to non-specialist residents, and being commited to inclusivity with regards to large-scale urban project which affect our lives as city dwellers. By being able to quantify and visualize advantages and shortcommings in design proposals, architects, developers and policy makers become better equipped to shape our cities for a better quality of life.

9. FUTURE DEVELOPMENT

While the present tool has been developed in to evalutate designs in an offline manner, new features could be developed in order to support online feedback on design proposal, the complication arises from the need to control and modify the geometry in the web browser, but with the advent of new CAD systems, this should be possible. This online feedback development would streamline urban analysis and boost public engagement. Inspired by the København Taler in Jernbanebyen initiative, the tool promotes collaborative planning, helping communities shape walkable, sustainable neighborhoods.Our upcoming platform, walkwisecopenhagen.com, will offer interactive walkability maps, making insights accessible to both professionals and the public.

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